O*NET OnLine

Database Administrators

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Administer, test, and implement computer databases, applying knowledge of database management systems. Coordinate changes to computer databases. Identify, investigate, and resolve database performance issues, database capacity, and database scalability. May plan, coordinate, and implement security measures to safeguard computer databases.

Sample of reported job titles: Database Administration Manager, Database Administrator (DBA), Database Analyst, Database Coordinator, Database Engineer, Database Manager, Development and Database Administration Manager, IS Admin (Information Systems Administrator), Systems Administrator (Systems Admin)

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

Occupation-Specific Information

Tasks

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Importance	Category	Task
75	Core	Modify existing databases and database management systems or direct programmers and analysts to make changes.
73	Core	Plan, coordinate, and implement security measures to safeguard information in computer files against accidental or unauthorized damage, modification or disclosure.
72	Core	Plan and install upgrades of database management system software to enhance database performance.

Importance	Category	Task
70	Core	Specify users and user access levels for each segment of database.
69	Core	Test changes to database applications or systems.
67	Core	Test programs or databases, correct errors, and make necessary modifications.
64	Core	Train users and answer questions.
62	Core	Provide technical support to junior staff or clients.
61	Core	Approve, schedule, plan, and supervise the installation and testing of new products and improvements to computer systems, such as the installation of new databases.
61	Core	Develop standards and guidelines for the use and acquisition of software and to protect vulnerable information.
54	Core	Write and code logical and physical database descriptions and specify identifiers of database to management system, or direct others in coding descriptions.
54	Core	Develop data models describing data elements and how they are used, following procedures and using pen, template, or computer software.
53	Core	Select and enter codes to monitor database performance and to create production databases.
51	Core	Identify, evaluate and recommend hardware or software technologies to achieve desired database performance.
51	Core	Review procedures in database management system manuals to make changes to database.
45	Supplemental	Identify and evaluate industry trends in database systems to serve as a source of information and advice for upper management.
45	Supplemental	Review workflow charts developed by programmer analyst to understand tasks computer will perform, such as updating records.

Importance	Category	Task
43	Supplemental	Revise company definition of data as defined in data dictionary.

Find occupations related to multiple tasks

Occupational Requirements

Work Activities

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Importance	Work Activity
96	**Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
80	**Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
78	**Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
78	**Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
75	**Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
73	**Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
72	**Establishing and Maintaining Interpersonal Relationships** — Developing constructive and cooperative working relationships with others, and maintaining them over time.
71	**Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.

Importance	Work Activity
69	**Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.
68	**Thinking Creatively** — Developing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.
67	**Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.
66	**Organizing, Planning, and Prioritizing Work** — Developing specific goals and plans to prioritize, organize, and accomplish your work.
65	**Documenting/Recording Information** — Entering, transcribing, recording, storing, or maintaining information in written or electronic/magnetic form.
62	**Interpreting the Meaning of Information for Others** — Translating or explaining what information means and how it can be used.
61	**Monitoring Processes, Materials, or Surroundings** — Monitoring and reviewing information from materials, events, or the environment, to detect or assess problems.
61	**Resolving Conflicts and Negotiating with Others** — Handling complaints, settling disputes, and resolving grievances and conflicts, or otherwise negotiating with others.
59	**Scheduling Work and Activities** — Scheduling events, programs, and activities, as well as the work of others.
58	**Developing Objectives and Strategies** — Establishing long-range objectives and specifying the strategies and actions to achieve them.
52	**Coordinating the Work and Activities of Others** — Getting members of a group to work together to accomplish tasks.
51	**Estimating the Quantifiable Characteristics of Products, Events, or Information** — Estimating sizes, distances, and quantities; or determining time, costs, resources, or materials needed to perform a work activity.
50	**Training and Teaching Others** — Identifying the educational needs of others, developing formal educational or training programs or classes, and teaching or instructing others.

Importance	Work Activity
48	**Developing and Building Teams** — Encouraging and building mutual trust, respect, and cooperation among team members.
48	**Providing Consultation and Advice to Others** — Providing guidance and expert advice to management or other groups on technical, systems-, or process-related topics.
46	**Assisting and Caring for Others** — Providing personal assistance, medical attention, emotional support, or other personal care to others such as coworkers, customers, or patients.
45	**Monitoring and Controlling Resources** — Monitoring and controlling resources and overseeing the spending of money.
42	**Judging the Qualities of Objects, Services, or People** — Assessing the value, importance, or quality of things or people.
40	**Communicating with People Outside the Organization** — Communicating with people outside the organization, representing the organization to customers, the public, government, and other external sources. This information can be exchanged in person, in writing, or by telephone or e-mail.
30	**Coaching and Developing Others** — Identifying the developmental needs of others and coaching, mentoring, or otherwise helping others to improve their knowledge or skills.
30	**Performing Administrative Activities** — Performing day-to-day administrative tasks such as maintaining information files and processing paperwork.
29	**Inspecting Equipment, Structures, or Materials** — Inspecting equipment, structures, or materials to identify the cause of errors or other problems or defects.
28	**Guiding, Directing, and Motivating Subordinates** — Providing guidance and direction to subordinates, including setting performance standards and monitoring performance.
25	**Repairing and Maintaining Electronic Equipment** — Servicing, repairing, calibrating, regulating, fine-tuning, or testing machines, devices, and equipment that operate primarily on the basis of electrical or electronic (not mechanical) principles.
23	**Staffing Organizational Units** — Recruiting, interviewing, selecting, hiring, and promoting employees in an organization.

Importance	Work Activity
22	**Controlling Machines and Processes** — Using either control mechanisms or direct physical activity to operate machines or processes (not including computers or vehicles).
20	**Selling or Influencing Others** — Convincing others to buy merchandise/goods or to otherwise change their minds or actions.
13	**Operating Vehicles, Mechanized Devices, or Equipment** — Running, maneuvering, navigating, or driving vehicles or mechanized equipment, such as forklifts, passenger vehicles, aircraft, or watercraft.
13	**Repairing and Maintaining Mechanical Equipment** — Servicing, repairing, adjusting, and testing machines, devices, moving parts, and equipment that operate primarily on the basis of mechanical (not electronic) principles.
12	**Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment** — Providing documentation, detailed instructions, drawings, or specifications to tell others about how devices, parts, equipment, or structures are to be fabricated, constructed, assembled, modified, maintained, or used.
11	**Performing General Physical Activities** — Performing general physical activities includes doing activities that require considerable use of your arms and legs and moving your whole body, such as climbing, lifting, balancing, walking, stooping, and handling materials.
10	**Handling and Moving Objects** — Using hands and arms in handling, installing, positioning, and moving materials, and manipulating things.
3	**Performing for or Working Directly with the Public** — Performing for people or dealing directly with the public. This includes serving customers in restaurants and stores, and receiving clients or guests.

Detailed Work Activities

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- Create databases to store electronic data.
- Update computer database information.
- Implement security measures for computer or information systems.
- Develop computer or information security policies or procedures.

- Install computer software.
- Assess database performance.
- Test computer system operations to ensure proper functioning.
- Modify software programs to improve performance.
- Train others in computer interface or software use.
- Provide technical support for software maintenance or use.
- Coordinate software or hardware installation.
- Develop detailed project plans.
- Develop performance metrics or standards related to information technology.
- Develop database parameters or specifications.
- Develop models of information or communications systems.
- Write computer programming code.
- Read documents to gather technical information.
- Evaluate utility of software or hardware technologies.
- Provide recommendations to others about computer hardware.
- Analyze data to identify trends or relationships among variables.
- Analyze market or customer related data.

Find occupations related to multiple detailed work activities

Work Context

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- **E-Mail** — How frequently does your job require you to use E-mail?

  92% Every day

- **Indoors, Environmentally Controlled** — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?

  90% Every day

- **Telephone Conversations** — How often do you have telephone conversations in this job?
- **Work With or Contribute to a Work Group or Team** — How important is it to work with or contribute to a work group or team in this job?
- **Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?

  56% A lot of freedom

  38% Some freedom

- **Importance of Being Exact or Accurate** — How important is being very exact or highly accurate in performing this job?
- **Spend Time Sitting** — How much does this job require sitting?

  24% More than half the time

- **Determine Tasks, Priorities and Goals** — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?
- **Contact With Others** — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?

  23% Constant contact with others

  71% Contact with others most of the time

- **Frequency of Decision Making** — How often is the worker required to make decisions that affect other people, the financial resources, and/or the image and reputation of the organization?

  17% Every day

  70% Once a week or more but not every day

- **Consequence of Error** — How serious would the result usually be if the worker made a mistake that was not easily correctable?
- **Spend Time Making Repetitive Motions** — How much does this job require making repetitive motions?

  18% About half the time

- **Coordinate or Lead Others in Accomplishing Work Activities** — How important is it to coordinate or lead others (not as a supervisor or team leader) in accomplishing work activities in this job?
- **Work Outcomes and Results of Other Workers** — How responsible is the worker for work outcomes and results of other workers?

  13% Limited responsibility

- **Importance of Repeating Same Tasks** — How important are continuous, repetitive, physical activities (like key entry) or mental activities (like checking entries in a ledger) to performing this job?

  17% Extremely important

  25% Important

- **Impact of Decisions on Co-workers or Company Results** — What results do your decisions usually have on other people or the image or reputation or financial resources of your employer?

  16% Very important results

  16% Moderate results

- **Face-to-Face Discussions with Individuals and Within Teams** — How frequently does your job require face-to-face discussions with individuals and within teams?

  16% Once a week or more but not every day

  15% Once a month or more but not every week

- **Duration of Typical Work Week** — Number of hours typically worked in one week.

16% More than 40 hours

68% 40 hours

15% Less than 40 hours

- **Spend Time Standing** — How much does this job require standing?

  26% Less than half the time

  13% Never

- **Time Pressure** — How often does this job require the worker to meet strict deadlines?

  22% Once a week or more but not every day

- **Written Letters and Memos** — How frequently does your job require written letters and memos?

  20% Once a year or more but not every month

- **Level of Competition** — To what extent does this job require the worker to compete or to be aware of competitive pressures?

  27% Highly competitive

- **Spend Time Using Your Hands to Handle, Control, or Feel Objects, Tools, or Controls** — How much does this job require using your hands to handle, control, or feel objects, tools or controls?
- **Degree of Automation** — How automated is the job?

  13% Highly automated

- **Conflict Situations** — How frequently are there conflict situations the employee has to face in this job?
- **Physical Proximity** — To what extent does this job require the worker to perform job tasks physically close to other people?

  12% Moderately close (at arm's length)

  21% I work with others but not closely (e.g., private office)

- **Dealing With Unpleasant, Angry, or Discourteous People** — How frequently does the worker have to deal with unpleasant, angry, or discourteous individuals as part of the job requirements?

  13% Once a week or more but not every day

  23% Once a year or more but not every month

- **Deal With External Customers or the Public in General** — How important is it to deal with external customers (as in retail sales) or the public in general (as in police work) in this job?
- **Exposed to Sounds, Noise Levels that are Distracting or Uncomfortable** — How often does this job require working exposed to sounds and noise levels that are distracting or uncomfortable?

  17% Once a year or more but not every month

- **Health and Safety of Other Workers** — How much responsibility is there for the health and safety of others in this job?

  20% High responsibility

- **Spend Time Bending or Twisting Your Body** — How much does this job require bending or twisting your body?

47% Never

- **In an Enclosed Vehicle or Operate Enclosed Equipment** — How often does this job require working in a closed vehicle or operate enclosed equipment (like a car)?
- **Public Speaking** — How frequently does your job require public speaking (one speaker with an audience)?
- **Wear Common Protective or Safety Equipment such as Safety Shoes, Glasses, Gloves, Hearing Protection, Hard Hats, or Life Jackets** — How often does this job require wearing common protective or safety equipment such as safety shoes, glasses, gloves, hearing protection, hard hats or life-jackets?
- **Spend Time Kneeling, Crouching, Stooping, or Crawling** — How much does this job require kneeling, crouching, stooping or crawling?

  14% Less than half the time

- **Spend Time Walking or Running** — How much does this job require walking or running?
- **Indoors, Not Environmentally Controlled** — How often does this job require working in an environment that is not environmentally controlled (like a warehouse without air conditioning)?

  15% Once a year or more but not every month

  75% Never

- **Dealing with Violent or Physically Aggressive People** — How frequently does this job require the worker to deal with physical aggression of violent individuals?

  12% Once a month or more but not every week

  88% Never

- **Pace Determined by Speed of Equipment** — How important is it to this job that the pace is determined by the speed of equipment or machinery? (This does not refer to keeping busy at all times on this job.)

  90% Not important at all

- **Exposed to High Places** — How often does this job require exposure to high places?

  90% Never

- **Work Schedules** — How regular are the work schedules for this job?

  93% Regular (established routine, set schedule)

- **Exposed to Contaminants** — How often does this job require working exposed to contaminants (such as pollutants, gases, dust or odors)?

  93% Never

- **Outdoors, Exposed to All Weather Conditions** — How often does this job require working outdoors, exposed to all weather conditions?

  12% Once a year or more but not every month

  88% Never

- **Outdoors, Under Cover** — How often does this job require working outdoors, under cover (like in an open shed)?

11% Once a year or more but not every month

88% Never

- **Exposed to Cramped Work Space, Awkward Positions** — How often does this job require working in cramped work spaces that requires getting into awkward positions?

  93% Never

- **Exposed to Extremely Bright or Inadequate Lighting Conditions** — How often does this job require working in extremely bright or inadequate lighting conditions?

  93% Never

- **Exposed to Hazardous Equipment** — How often does this job require exposure to hazardous equipment?

  94% Never

- **Exposed to Very Hot or Cold Temperatures** — How often does this job require working in very hot (above 90 F degrees) or very cold (below 32 F degrees) temperatures?

  92% Never

- **Spend Time Climbing Ladders, Scaffolds, or Poles** — How much does this job require climbing ladders, scaffolds, poles, etc.?

  93% Never

- **Wear Specialized Protective or Safety Equipment such as Breathing Apparatus, Safety Harness, Full Protection Suits, or Radiation Protection** — How often does this job require wearing specialized protective or safety equipment such as breathing apparatus, safety harness, full protection suits, or radiation protection?

  93% Never

- **Exposed to Disease or Infections** — How often does this job require exposure to disease/infections?

  95% Never

- **Exposed to Hazardous Conditions** — How often does this job require exposure to hazardous conditions?

  95% Never

- **In an Open Vehicle or Operating Equipment** — How often does this job require working in an open vehicle or operating equipment (like a tractor)?

  98% Never

- **Spend Time Keeping or Regaining Balance** — How much does this job require keeping or regaining your balance?

  95% Never

- **Exposed to Minor Burns, Cuts, Bites, or Stings** — How often does this job require exposure to minor burns, cuts, bites, or stings?

  100% Never

Exposed to Radiation — How often does this job require exposure to radiation?

100% Never

Exposed to Whole Body Vibration — How often does this job require exposure to whole body vibration (like operating a jackhammer or earth moving equipment)?

100% Never

Experience Requirements

Job Zone

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Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators.
SVP Range	(7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

Apprenticeship Opportunities

All 2 displayed

Example apprenticeship titles for this occupation:

- Database Administrator (Nof)
- Database Technician

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Access software** — Access management software; Citrix cloud computing software
- **Accounting software** — Fund accounting software
- **Administration software** — Redgate SQL Server
- **Analytical or scientific software** — IBM SPSS Statistics; Minitab; SAS; The MathWorks MATLAB; 3 more
- **Application server software** — Docker; Kubernetes; Red Hat OpenShift; Spring Boot; 3 more
- **Backup or archival software** — Acronis Recovery Expert; EMC NetWorker; HP DataProtector; Veritas NetBackup; 6 more
- **Business intelligence and data analysis software** — Apache Spark; IBM Cognos Impromptu; Microsoft Power BI; Tableau; 4 more
- **Cloud-based management software** — Amazon Web Services AWS CloudFormation; IBM WebSphere; Microsoft Azure Data Factory; Splunk Enterprise
- **Cloud-based protection or security software** — SolarWinds
- **Clustering software** — Cluster server software; Oracle Real Application Cluster RAC; VMware
- **Communications server software** — IBM Domino
- **Computer aided design CAD software** — Autodesk Revit; Dassault Systemes CATIA
- **Configuration management software** — Chef; IBM Terraform; Perforce Helix software; Puppet; 2 more
- **Content workflow software** — Atlassian JIRA
- **Customer relationship management CRM software** — Blackbaud The Raiser's Edge; Oracle Eloqua; Salesforce software
- **Data base management system software** — Amazon DynamoDB; Elasticsearch; MongoDB; Oracle PL/SQL; 35 more

- **Data base reporting software** — Microsoft SQL Server Reporting Services SSRS; Oracle Reports; Oracle SQL Plus; SAP Crystal Reports; 3 more
- **Data base user interface and query software** — Apache Hive; Blackboard software; IBM DB2; MySQL; 12 more
- **Data mining software** — Google Analytics; Snowflake
- **Desktop communications software** — Skype
- **Desktop publishing software** — Microsoft Publisher
- **Development environment software** — Apache Kafka; Apache Maven; Go; Oracle SQL Developer; 15 more
- **Document management software** — Microsoft SharePoint
- **Electronic mail software** — IBM Notes; Microsoft Exchange; Microsoft Outlook
- **Enterprise application integration software** — BMC Software Control-M; Extensible markup language XML; IBM InfoSphere DataStage; Microsoft SQL Server Integration Services SSIS; 2 more
- **Enterprise resource planning ERP software** — Ellucian Banner ERP; Microsoft Dynamics; Oracle PeopleSoft; SAP software; 9 more
- **Enterprise system management software** — IBM Power Systems software
- **Expert system software** — Ansible software; Apache Mahout
- **File versioning software** — Apache Subversion SVN; Git
- **Financial analysis software** — Delphi Technology; Oracle E-Business Suite Financials
- **Geographic information system** — ESRI ArcGIS software; Geographic information system GIS software
- **Graphical user interface development software** — Salesforce Visualforce
- **Human resources software** — Human resource management software HRMS; Oracle Learning Management; Oracle Taleo
- **Industrial control software** — Supervisory control and data acquisition SCADA software
- **Information retrieval or search software** — Data validation software; LexisNexis
- **Manufacturing execution system MES software** — CA Easytrieve Report Generator
- **Medical software** — Epic Systems
- **Metadata management software** — IBM Rational Data Architect; Informatica software; Pentaho Kettle; Quest Erwin Data Modeler; 3 more
- **Multi-media educational software** — Nearpod
- **Network conferencing software** — LogMeIn GoToWebinar
- **Network monitoring software** — Nagios; Wireshark; Zabbix
- **Network security and virtual private network VPN equipment software** — Virtual private networking VPN software
- **Network security or virtual private network VPN management software** — Database security software
- **Object or component oriented development software** — C#; jQuery; Scala; Swift; 12 more
- **Object oriented data base management software** — Hibernate ORM; IBM Informix; PostgreSQL; Transact-SQL; 1 more

- **Office suite software** — Microsoft Office software
- **Operating system software** — Bash ; Microsoft Windows Server ; Red Hat Enterprise Linux ; UNIX Shell ; 11 more
- **Portal server software** — Apache HTTP Server
- **Presentation software** — Microsoft PowerPoint
- **Procedure management software** — Apache Airflow
- **Process mapping and design software** — Microsoft Visio ; Visual Paradigm DB Visual ARCHITECT
- **Program testing software** — Database testing software; Hewlett Packard LoadRunner; JUnit ; Selenium ; 1 more
- **Project management software** — Atlassian Confluence ; Microsoft Project ; Oracle Primavera Enterprise Project Portfolio Management
- **Requirements analysis and system architecture software** — Database capacity planning software; Unified modeling language UML
- **Sales and marketing software** — Marketo Marketing Automation
- **Spreadsheet software** — Microsoft Excel
- **Storage networking software** — Amazon Simple Storage Service S3; Storage area network SAN software
- **Transaction security and virus protection software** — Encryption software; McAfee; NortonLifeLock cybersecurity software
- **Transaction server software** — Customer information control system CICS
- **Video conferencing software** — Cisco Webex ; Google Meet; LogMeIn GoToMeeting
- **Web page creation and editing software** — Adobe Dreamweaver
- **Web platform development software** — Django ; Google Angular ; Microsoft ASP.NET ; Spring Framework ; 22 more
- **Word processing software** — Microsoft Word

Hot Technologies are requirements most frequently included across all employer job postings.
See all 112 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 38 In Demand skills for this occupation.

Essential Skills

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Importance	Essential Skill
72	**Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
69	**Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and

Importance	Essential Skill
	not interrupting at inappropriate times.
69	**Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
63	**Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
60	**Speaking** — Talking to others to convey information effectively.
56	**Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
56	**Writing** — Communicating effectively in writing as appropriate for the needs of the audience.
50	**Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
50	**Mathematics** — Using mathematics to solve problems.
25	**Science** — Using scientific rules and methods to solve problems.

Transferable Skills

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Importance	Transferable Skill
72	**Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
69	**Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
60	**Programming** — Writing computer programs for various purposes.
53	**Operations Analysis** — Analyzing needs and product requirements to create a design.

Importance	Transferable Skill
53	**Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
53	**Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
50	**Coordination** — Adjusting actions in relation to others' actions.
50	**Instructing** — Teaching others how to do something.
47	**Service Orientation** — Actively looking for ways to help people.
47	**Social Perceptiveness** — Being aware of others' reactions and understanding why they react as they do.
47	**Time Management** — Managing one's own time and the time of others.
44	**Management of Personnel Resources** — Motivating, developing, and directing people as they work, identifying the best people for the job.
44	**Operations Monitoring** — Watching gauges, dials, or other indicators to make sure a machine is working properly.
41	**Persuasion** — Persuading others to change their minds or behavior.
38	**Quality Control Analysis** — Conducting tests and inspections of products, services, or processes to evaluate quality or performance.
38	**Troubleshooting** — Determining causes of operating errors and deciding what to do about it.
35	**Technology Design** — Generating or adapting equipment and technology to serve user needs.
25	**Negotiation** — Bringing others together and trying to reconcile differences.
22	**Management of Material Resources** — Obtaining and seeing to the appropriate use of equipment, facilities, and materials needed to do certain work.
19	**Management of Financial Resources** — Determining how money will be spent to get the work done, and accounting for these expenditures.

Importance	Transferable Skill
13	**Equipment Selection** — Determining the kind of tools and equipment needed to do a job.
13	**Operation and Control** — Controlling operations of equipment or systems.
10	**Installation** — Installing equipment, machines, wiring, or programs to meet specifications.
3	**Repairing** — Repairing machines or systems using the needed tools.
0	**Equipment Maintenance** — Performing routine maintenance on equipment and determining when and what kind of maintenance is needed.

Knowledge

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Importance	Knowledge
95	**Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
74	**English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
73	**Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
64	**Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
64	**Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
63	**Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles,

Importance	Knowledge
	techniques, procedures, and equipment to the design and production of various goods and services.
53	**Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.
52	**Education and Training** — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.
45	**Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.
43	**Economics and Accounting** — Knowledge of economic and accounting principles and practices, the financial markets, banking, and the analysis and reporting of financial data.
39	**Communications and Media** — Knowledge of media production, communication, and dissemination techniques and methods. This includes alternative ways to inform and entertain via written, oral, and visual media.
39	**Public Safety and Security** — Knowledge of relevant equipment, policies, procedures, and strategies to promote effective local, state, or national security operations for the protection of people, data, property, and institutions.
38	**Personnel and Human Resources** — Knowledge of principles and procedures for personnel recruitment, selection, training, compensation and benefits, labor relations and negotiation, and personnel information systems.
36	**Law and Government** — Knowledge of laws, legal codes, court procedures, precedents, government regulations, executive orders, agency rules, and the democratic political process.
34	**Administrative** — Knowledge of administrative and office procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and workplace terminology.
28	**Production and Processing** — Knowledge of raw materials, production processes, quality control, costs, and other techniques for maximizing the effective manufacture and distribution of goods.

Importance	Knowledge
25	**Transportation** — Knowledge of principles and methods for moving people or goods by air, rail, sea, or road, including the relative costs and benefits.
22	**Psychology** — Knowledge of human behavior and performance; individual differences in ability, personality, and interests; learning and motivation; psychological research methods; and the assessment and treatment of behavioral and affective disorders.
18	**Sales and Marketing** — Knowledge of principles and methods for showing, promoting, and selling products or services. This includes marketing strategy and tactics, product demonstration, sales techniques, and sales control systems.
17	**Mechanical** — Knowledge of machines and tools, including their designs, uses, repair, and maintenance.
17	**Medicine and Dentistry** — Knowledge of the information and techniques needed to diagnose and treat human injuries, diseases, and deformities. This includes symptoms, treatment alternatives, drug properties and interactions, and preventive health-care measures.
15	**Physics** — Knowledge and prediction of physical principles, laws, their interrelationships, and applications to understanding fluid, material, and atmospheric dynamics, and mechanical, electrical, atomic and sub-atomic structures and processes.
13	**Geography** — Knowledge of principles and methods for describing the features of land, sea, and air masses, including their physical characteristics, locations, interrelationships, and distribution of plant, animal, and human life.
9	**Foreign Language** — Knowledge of the structure and content of a foreign (non-English) language including the meaning and spelling of words, rules of composition and grammar, and pronunciation.
8	**Building and Construction** — Knowledge of materials, methods, and the tools involved in the construction or repair of houses, buildings, or other structures such as highways and roads.
7	**Chemistry** — Knowledge of the chemical composition, structure, and properties of substances and of the chemical processes and transformations that they undergo. This includes uses of chemicals and their interactions, danger signs, production techniques, and disposal methods.
7	**History and Archeology** — Knowledge of historical events and their causes, indicators, and effects on civilizations and cultures.

Importance	Knowledge
5	**Sociology and Anthropology** — Knowledge of group behavior and dynamics, societal trends and influences, human migrations, ethnicity, cultures, and their history and origins.
4	**Biology** — Knowledge of plant and animal organisms, their tissues, cells, functions, interdependencies, and interactions with each other and the environment.
3	**Philosophy and Theology** — Knowledge of different philosophical systems and religions. This includes their basic principles, values, ethics, ways of thinking, customs, practices, and their impact on human culture.
2	**Therapy and Counseling** — Knowledge of principles, methods, and procedures for diagnosis, treatment, and rehabilitation of physical and mental dysfunctions, and for career counseling and guidance.
1	**Fine Arts** — Knowledge of the theory and techniques required to compose, produce, and perform works of music, dance, visual arts, drama, and sculpture.
1	**Food Production** — Knowledge of techniques and equipment for planting, growing, and harvesting food products (both plant and animal) for consumption, including storage/handling techniques.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

89%	Bachelor's degree required
4%	Post-baccalaureate certificate required
3%	Associate's degree or other 2-year degree required

Worker Characteristics

Abilities

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Importance	Ability
75	**Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
72	**Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
72	**Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
72	**Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
72	**Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
72	**Written Comprehension** — The ability to read and understand information and ideas presented in writing.
66	**Near Vision** — The ability to see details at close range (within a few feet of the observer).
63	**Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
60	**Category Flexibility** — The ability to generate or use different sets of rules for combining or grouping things in different ways.
60	**Written Expression** — The ability to communicate information and ideas in writing so others will understand.
56	**Flexibility of Closure** — The ability to identify or detect a known pattern (a figure, object, word, or sound) that is hidden in other distracting material.
56	**Speech Recognition** — The ability to identify and understand the speech of another person.
53	**Fluency of Ideas** — The ability to come up with a number of ideas about a topic (the number of ideas is important, not their quality, correctness, or creativity).
53	**Mathematical Reasoning** — The ability to choose the right mathematical methods or formulas to solve a problem.

Importance	Ability
53	**Speech Clarity** — The ability to speak clearly so others can understand you.
50	**Number Facility** — The ability to add, subtract, multiply, or divide quickly and correctly.
50	**Originality** — The ability to come up with unusual or clever ideas about a given topic or situation, or to develop creative ways to solve a problem.
50	**Perceptual Speed** — The ability to quickly and accurately compare similarities and differences among sets of letters, numbers, objects, pictures, or patterns. The things to be compared may be presented at the same time or one after the other. This ability also includes comparing a presented object with a remembered object.
50	**Selective Attention** — The ability to concentrate on a task over a period of time without being distracted.
50	**Visualization** — The ability to imagine how something will look after it is moved around or when its parts are moved or rearranged.
47	**Speed of Closure** — The ability to quickly make sense of, combine, and organize information into meaningful patterns.
41	**Far Vision** — The ability to see details at a distance.
38	**Memorization** — The ability to remember information such as words, numbers, pictures, and procedures.
31	**Time Sharing** — The ability to shift back and forth between two or more activities or sources of information (such as speech, sounds, touch, or other sources).
25	**Arm-Hand Steadiness** — The ability to keep your hand and arm steady while moving your arm or while holding your arm and hand in one position.
22	**Auditory Attention** — The ability to focus on a single source of sound in the presence of other distracting sounds.
22	**Control Precision** — The ability to quickly and repeatedly adjust the controls of a machine or a vehicle to exact positions.
22	**Visual Color Discrimination** — The ability to match or detect differences between colors, including shades of color and brightness.

Importance	Ability
19	**Finger Dexterity** — The ability to make precisely coordinated movements of the fingers of one or both hands to grasp, manipulate, or assemble very small objects.
19	**Manual Dexterity** — The ability to quickly move your hand, your hand together with your arm, or your two hands to grasp, manipulate, or assemble objects.
19	**Trunk Strength** — The ability to use your abdominal and lower back muscles to support part of the body repeatedly or continuously over time without "giving out" or fatiguing.
16	**Hearing Sensitivity** — The ability to detect or tell the differences between sounds that vary in pitch and loudness.
13	**Wrist-Finger Speed** — The ability to make fast, simple, repeated movements of the fingers, hands, and wrists.
10	**Depth Perception** — The ability to judge which of several objects is closer or farther away from you, or to judge the distance between you and an object.
10	**Response Orientation** — The ability to choose quickly between two or more movements in response to two or more different signals (lights, sounds, pictures). It includes the speed with which the correct response is started with the hand, foot, or other body part.
6	**Dynamic Strength** — The ability to exert muscle force repeatedly or continuously over time. This involves muscular endurance and resistance to muscle fatigue.
3	**Reaction Time** — The ability to quickly respond (with the hand, finger, or foot) to a signal (sound, light, picture) when it appears.
3	**Stamina** — The ability to exert yourself physically over long periods of time without getting winded or out of breath.
0	**Dynamic Flexibility** — The ability to quickly and repeatedly bend, stretch, twist, or reach out with your body, arms, and/or legs.
0	**Explosive Strength** — The ability to use short bursts of muscle force to propel oneself (as in jumping or sprinting), or to throw an object.
0	**Extent Flexibility** — The ability to bend, stretch, twist, or reach with your body, arms, and/or legs.

Importance	Ability
0	**Glare Sensitivity** — The ability to see objects in the presence of a glare or bright lighting.
0	**Gross Body Coordination** — The ability to coordinate the movement of your arms, legs, and torso together when the whole body is in motion.
0	**Gross Body Equilibrium** — The ability to keep or regain your body balance or stay upright when in an unstable position.
0	**Multilimb Coordination** — The ability to coordinate two or more limbs (for example, two arms, two legs, or one leg and one arm) while sitting, standing, or lying down. It does not involve performing the activities while the whole body is in motion.
0	**Night Vision** — The ability to see under low-light conditions.
0	**Peripheral Vision** — The ability to see objects or movement of objects to one's side when the eyes are looking ahead.
0	**Rate Control** — The ability to time your movements or the movement of a piece of equipment in anticipation of changes in the speed and/or direction of a moving object or scene.
0	**Sound Localization** — The ability to tell the direction from which a sound originated.
0	**Spatial Orientation** — The ability to know your location in relation to the environment or to know where other objects are in relation to you.
0	**Speed of Limb Movement** — The ability to quickly move the arms and legs.
0	**Static Strength** — The ability to exert maximum muscle force to lift, push, pull, or carry objects.

Career Interest Types

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Occupational Interest	Career Interest Type
94	**Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
64	**Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
32	**Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.
31	**Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
22	**Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.
3	**Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.

Specific Interest Areas

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Occupational Interest	Specific Interest Area
95	**Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
44	**Management/Administration** — Work involves planning, directing, or coordinating the daily operations of an organization, and overseeing personnel, business, or administrative services.
37	**Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.
26	**Office Work** — Work involves clerical and administrative tasks for an organization such as answering phones, bookkeeping, updating records, scheduling appointments or meetings, and editing of documents and files.
16	**Accounting** — Work involves examining, analyzing, and interpreting accounting records for financial statements, or maintaining, auditing, or evaluating records and accounts.
15	**Engineering** — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.
14	**Mechanics/Electronics** — Work involves maintaining and repairing of machinery, such as automotive engines, electronic equipment, computers, and communication systems.
13	**Teaching/Education** — Work involves teaching through lectures, discussions, and demonstrations in one or more subjects.
11	**Human Resources** — Work involves recruiting, selecting, and training job applicants, maintaining employee personnel records, administering compensation and employee-benefit programs, and conducting employee performance reviews.
9	**Business Initiatives** — Work involves creating strategic plans or taking actions to expand an organization's business or market share.
8	**Finance** — Work involves directing and managing financial activities, such as planning and investments for an individual or organization to generate income.
8	**Public Speaking** — Work involves speaking before an audience to educate, entertain, or influence the listeners.

Occupational Interest	Specific Interest Area
6	**Personal Service** — Work involves providing personal services to individuals in need of non-medical assistance such as helping arrange or plan for travel, meals, or events.
6	**Professional Advising** — Work involves providing expert advice, coaching, and counseling on topics such as careers, educational opportunities, and educational and professional development.
6	**Protective Service** — Work involves maintaining order and protecting life and property by conducting investigations and enforcing laws and regulations to prevent or solve crimes.
5	**Physical Science** — Work involves studying and researching non-living objects, materials, and phenomena through scientific disciplines such as physics, chemistry, astronomy, meteorology, or geology.
4	**Health Care Service** — Work involves providing health care services to individuals to include the diagnosis, prevention, or treatment of diseases and other forms of illness or impairment.
4	**Law** — Work involves studying and applying legal knowledge.
4	**Medical Science** — Work involves studying and researching the causes, nature, effects, and treatments of disease or other forms of impairment or illness.
3	**Social Science** — Work involves studying and researching human behavior, society, politics, and social relationships through scientific disciplines such as psychology, sociology, economics, or political science.
2	**Life Science** — Work involves studying and researching living organisms through scientific disciplines such as biology, botany, zoology, microbiology, physiology, biochemistry, or zoology.
2	**Marketing/Advertising** — Work involves promoting or creating a public image for products, services, or organizations to attract interest, engagement, or sales.
2	**Media** — Work involves producing, developing, writing, directing, or presenting information through print, radio, television, or digital media.
2	**Sales** — Work involves selling goods or services.
1	**Creative Writing** — Work involves developing creative written works, such as novels, scripts, song lyrics, poetry, stories, or advertisements.

Occupational Interest	Specific Interest Area
1	**Politics** — Work involves engaging in political activities to influence government policy.
1	**Social Service** — Work involves providing social services and assistance to improve the well-being and social and psychological functioning of children and adults.
0	**Agriculture** — Work involves planting, cultivating, and harvesting crops, plants, or trees, or raising farm animals. Work can also involve preparing plant and animal products for distribution to consumers, or protecting natural resources to maximize their use in farming.
0	**Animal Service** — Work involves feeding, grooming, exercising, or otherwise providing care to promote and maintain the well-being of pets and other animals in settings such as kennels, animal shelters, and zoos.
0	**Applied Arts and Design** — Work involves applying artistic principles to the design and creation of various objects and materials.
0	**Athletics** — Work involves training, practicing, or competing in athletic events, or supporting those participating in athletics through coaching, training, or officiating.
0	**Construction/Woodwork** — Work involves constructing, installing, or repairing structures and fixtures made of wood.
0	**Culinary Art** — Work involves preparing, cooking, and presenting food in a distinctive style usually associated with a specific culture or region.
0	**Humanities** — Work involves studying and researching aspects of human history, culture, and language through disciplines such as history, philosophy, the language arts, or area/ethnic studies.
0	**Music** — Work involves conducting, composing, or playing music or singing.
0	**Nature/Outdoors** — Work involves planning, restoring, and conserving natural areas and wildlife habitats.
0	**Performing Arts** — Work involves acting as a performer in dance, comedy, or dramatic productions, choreographing of dance performances, or producing of live or recorded entertainment.

Occupational Interest	Specific Interest Area
0	**Physical/Manual Labor** — Work involves manually unloading and moving freight, stock, luggage, or other materials, or performing other physical labor.
0	**Religious Activities** — Work involves conducting religious worship, leading a church or other religious group, and offering moral guidance associated with beliefs and practices of a religious faith.
0	**Transportation/Machine Operation** — Work involves driving vehicles, such as vans, trucks, trains, and buses to transport people or materials. Work can also involve operating heavy equipment, such as cranes, forklifts, bulldozers, and backhoes to move materials or earth.
0	**Visual Arts** — Work involves creating original visual artwork for a variety of media and purposes using various techniques and software.

Work Styles

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Impact	Work Style
100	**Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
88	**Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
82	**Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
77	**Integrity** — A tendency to be honest and ethical at work.
73	**Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
59	**Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.

Impact	Work Style
57	**Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
56	**Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
49	**Initiative** — A tendency to be proactive and take on extra responsibilities and tasks that may fall outside of one's required work role.
44	**Tolerance for Ambiguity** — A tendency to be comfortable with ambiguity and uncertainty at work.
41	**Innovation** — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work.
40	**Cooperation** — A tendency to be pleasant, helpful, and willing to assist others at work.
40	**Stress Tolerance** — A tendency to cope and function effectively in stressful situations at work.
38	**Self-Confidence** — A tendency to believe in one's work-related capabilities and ability to control one's work-related outcomes.
38	**Self-Control** — A tendency to remain calm and composed and to manage emotions effectively in response to criticism or difficult situations at work.
30	**Leadership Orientation** — A tendency to lead, take charge, offer opinions, and provide direction at work.
18	**Sincerity** — A tendency to be genuine and sincere in interactions with others at work, without concern for personal gain or self-interest.
11	**Humility** — A tendency to be modest and humble when interacting with others at work.
9	**Social Orientation** — A tendency to seek out, enjoy, and be energized by social interaction at work.
4	**Optimism** — A tendency to exhibit a positive attitude and positive emotions at work, even under difficult circumstances.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025) $50.30 hourly, $104,620 annual

State wages Select a State Go

Local wages ZIP Code Go

Employment (2025) 75,000 employees

Projected growth (2025-2035) Little or no change

Projected job openings (2025-2035) 3,300

State trends Select a State Go

Top industries (2025)
Professional, Scientific, and Technical Services (23% employed in this sector)
Finance and Insurance (12%)
Educational Services (11%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 15-1211.00 Computer Systems Analysts Bright Outlook
- 15-1299.08 Computer Systems Engineers/Architects

- 15-1243.01 Data Warehousing Specialists
- 15-1243.00 Database Architects
- 15-1299.03 Document Management Specialists
- 15-1212.00 Information Security Analysts
- 15-1299.05 Information Security Engineers
- 15-1244.00 Network and Computer Systems Administrators
- 15-1252.00 Software Developers
- 15-1299.01 Web Administrators

Supplemental related occupations

- 15-1251.00 Computer Programmers
- 15-1299.07 Blockchain Engineers
- 15-1253.00 Software Quality Assurance Analysts and Testers
- 15-1231.00 Computer Network Support Specialists
- 15-1255.00 Web and Digital Interface Designers
- 15-1241.00 Computer Network Architects
- 15-1232.00 Computer User Support Specialists
- 11-3021.00 Computer and Information Systems Managers
- 15-1254.00 Web Developers
- 15-1299.04 Penetration Testers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

- Association for Computing Machinery
- Computing Research Association
- IEEE Computer Society
- National Center for Women and Information Technology

Accreditation, Certification, & Unions

- CompTIA
- Institute for Certification of Computing Professionals